The Diagnostic Penumbra: Constructive Reverse Mathematics of Bayesian Clinical Decision-Making

Paper 104, Clinical Sub-Series Paper B,
of the Constructive Reverse Mathematics Series

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Abstract

Every diagnostic test produces a finite contingency table — true positives, false positives, false negatives, true negatives — from which sensitivity, specificity, and predictive values are computed as rational numbers (BISH). The clinical decision invokes Bayes' theorem: the posterior probability $P(D \mid +)$ is compared against a treatment threshold τ to determine whether to treat. When prevalence is a rational cohort statistic, the posterior is rational and the decision is BISH-decidable. When prevalence is a real-valued population parameter, the posterior passes through the continuum, and comparison with τ requires Markov's Principle (BISH + MP) — structurally identical to the p-value comparison in Paper 103. We identify the *Diagnostic Penumbra*: the grey zone where the posterior lies within δ of the treatment threshold, where δ combines prevalence uncertainty and analytical imprecision. For ROC curve analysis, comparing AUCs of two tests requires LPO (integral over continuous ROC); discrete trapezoidal AUC is BISH. The treatment threshold from continuous utility functions requires WKL (bounded optimisation).

The diagnostic pipeline decomposes as 7 BISH+1 BISH+MP+1 WKL+1 LPO = 10 stages (70% BISH) — isomorphic to Paper 103. Applied to the ESC 0/1h high-sensitivity troponin algorithm: 24.7% of chest pain presentations (1,972 of 7,998 in the Mueller validation cohort) fall in the grey zone (5–52 ng/L), with an event rate of 7.1% that sits squarely in the treatment-threshold range for invasive investigation. Applied to D-dimer for PE exclusion and PSA screening, the same structure recurs: rational data is BISH; the grey zone emerges at the interface with the continuum. All results are formalised in Lean 4 with Mathlib (10 module files, 0 sorries on main theorems, 4 documentary axioms). This is the second paper in the CRM Clinical Sub-Series.

1 Clinical Premise

A cardiologist receives a 58-year-old man in the emergency department with acute chest pain. The ECG is non-diagnostic. The first high-sensitivity troponin result returns at 12 ng/L — above the rule-out threshold of 5 ng/L but below the rule-in threshold of 52 ng/L. The patient is in the *observe zone*.

The clinician must decide: admit for serial troponins and possible angiography, or discharge with outpatient follow-up? The decision depends on the posterior probability of acute coronary syndrome given the test result, compared against the treatment threshold determined by the relative costs of missed diagnosis versus unnecessary catheterisation.

Three questions arise:

1. **What does "in the grey zone" actually mean?** The contingency table data and the test characteristics are rational numbers, computable from finite counts. But the population prevalence of ACS in chest pain presentations is a real-valued parameter estimated from epidemiological studies. The posterior probability inherits this real-valuedness. Comparing a real-valued posterior with a rational threshold is not a finite computation.
2. **Is the grey zone a statistical artefact or a logical structure?** The ESC 0/1h algorithm's observe zone (5–52 ng/L) captures 24.7% of patients. We show this zone is not merely a practical inconvenience but a *logical structure* — the Diagnostic Penumbra — where the decision is classically resolvable but not constructively certifiable.
3. **When does the grey zone collapse?** With rational cohort prevalence (a specific study population), the posterior is rational and the decision is decidable in BISH. The grey zone is a property of the passage from finite data to the continuum, not of the data itself.

These questions are the diagnostic analogues of the questions Paper 103 posed for the RCT statistical pipeline. There, the transcendental gate was the normal CDF Φ; here, it is Bayes' theorem with real prevalence. The mechanism is the same: rational data passes through a real-valued function, producing a quantity whose comparison with a rational threshold requires omniscience.

2 Introduction

2.1 Main Results

Theorem 2.1 (A: Diagnostic Penumbra). *A diagnostic result with posterior $P(D \mid +)$ and total diagnostic uncertainty $\delta = \delta_{\text{prev}} + \delta_{\text{cv}}$ lies in the* Diagnostic Penumbra *at treatment threshold τ if and only if*

$$|P(D \mid +) - \tau| \ < \ \delta.$$

The penumbra has width 2δ, symmetric around τ. Within this band, the treat/no-treat decision is classically resolvable but not constructively certifiable.

Theorem 2.2 (B: Bayesian MP Requirement). *When the prevalence π is a real-valued population parameter, the posterior*

$$P(D \mid +) = \frac{\text{sens} \cdot \pi}{\text{sens} \cdot \pi + (1 - \text{spec})(1 - \pi)}$$

is a computable real. Comparing $P(D \mid +)$ with a rational threshold τ requires BISH+MP. This is structurally identical to Paper 103's Theorem B: rational data $\to$ transcendental function $\to$ real comparison $\to$ Markov's Principle.

Theorem 2.3 (C: ROC = LPO). *Comparing AUCs of two diagnostic tests, where each AUC is the Riemann integral $\int_0^1 \text{ROC}(t)\,dt$ over the continuous false-positive-rate axis, requires LPO (universal real comparison). With discrete trapezoidal AUC from finitely many threshold points, the comparison is rational and BISH-decidable.*

Theorem 2.4 (D: Discrete Bypass). *When the prevalence comes from a finite cohort study (rational), the entire Bayesian chain is BISH:*

$$\text{sens}, \text{spec}, \pi_{\text{cohort}} \in \mathbb{Q} \implies P(D \mid +) \in \mathbb{Q} \implies \text{“}P(D \mid +) > \tau\text{?”} \text{ decidable in BISH.}$$

This is the diagnostic analogue of Fisher's exact test in Paper 103: stay in $\mathbb{Q}$, pay no logical cost.

2.2 CRM Primer

Constructive Reverse Mathematics classifies mathematical theorems by the weakest axiom system above BISH (Bishop's constructive mathematics) required to prove them. The hierarchy

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}$$

measures *logical cost*: how much omniscience is needed. Papers 1–102 established that the logical cost of mathematics is the logical cost of $\mathbb{R}$ — the completeness and total ordering at the Archimedean place [16]. Paper 103 [17] opened the Clinical Sub-Series by applying CRM to the RCT statistical pipeline. Paper 104 extends the framework to diagnostic testing.

2.3 Atlas Fit

Within the CRM atlas, Paper 104 demonstrates the same *Archimedean obstruction* as Paper 103 in a structurally distinct clinical domain:

- **P103:** data $\xrightarrow{\text{CLT}} \Phi \xrightarrow{\text{compare}} p < \alpha$ — the transcendental gate is the normal CDF.
- **P104:** data $\xrightarrow{\text{Bayes}} P(D \mid +) \xrightarrow{\text{compare}} P > \tau$ — the transcendental gate is Bayesian inference with real prevalence.

Both produce the same CRM decomposition: 7 BISH + 1 BISH+MP + 1 WKL + 1 LPO = 10 stages (70% BISH). The 70% coincidence reflects the universal structure of the obstruction.

3 Preliminaries

3.1 Omniscience Principles

We follow Bishop–Bridges [1] and Troelstra–van Dalen [2]. A *binary sequence* is a function $\alpha : \mathbb{N} \to \{0, 1\}$.

Definition 3.1 (LPO)**.** The *Limited Principle of Omniscience*: for every binary sequence α, either $\exists\, n\, (\alpha(n) = 1)$ or $\forall\, n\, (\alpha(n) = 0)$.

Definition 3.2 (MP)**.** *Markov's Principle*: if $\neg\forall\, n\, (\alpha(n) = 0)$, then $\exists\, n\, (\alpha(n) = 1)$.

Definition 3.3 (WLPO)**.** The *Weak Limited Principle of Omniscience*: for every α, either $\forall\, n\, (\alpha(n) = 0)$ or $\neg\forall\, n\, (\alpha(n) = 0)$.

3.2 Decidability Hierarchy

- **Level 0 (BISH):** Comparison of rationals $p, q \in \mathbb{Q}$ is decidable.
- **Level 1 (BISH + MP):** Comparison of a computable real with a rational, given a $\neg\neg$-witness, terminates by Markov's Principle.
- **Level 2 (LPO):** Universal trichotomy on $\mathbb{R}$ is equivalent to LPO.

3.3 Bayes' Theorem

For a diagnostic test with sensitivity sens and specificity spec applied to a population with disease prevalence π:

$$P(D \mid +) = \frac{\text{sens} \cdot \pi}{\text{sens} \cdot \pi + (1 - \text{spec})(1 - \pi)}. \tag{1}$$

When $\text{sens}, \text{spec}, \pi \in \mathbb{Q}$, the posterior $P(D \mid +) \in \mathbb{Q}$. When $\pi \in \mathbb{R} \setminus \mathbb{Q}$, the posterior inherits real-valuedness.

3.4 Treatment Threshold

The Pauker–Kassirer threshold [3] for a treat/no-treat decision:

$$\tau = \frac{C_{\mathrm{Rx}}}{B_{\mathrm{Rx}} + C_{\mathrm{Rx}}} \tag{2}$$

where B_{Rx} is the net benefit of treating true disease and C_{Rx} is the net cost of treating non-disease. With rational utilities, $\tau \in \mathbb{Q}$.

4 Main Results

4.1 The Contingency Table Is BISH

Proposition 4.1. *Every 2×2 contingency table $(TP, FP, FN, TN) \in \mathbb{N}^4$ produces rational test characteristics:*

$$\mathrm{sens} = \frac{TP}{TP + FN} \in \mathbb{Q}, \quad \mathrm{spec} = \frac{TN}{TN + FP} \in \mathbb{Q}, \quad PPV = \frac{TP}{TP + FP} \in \mathbb{Q}, \quad NPV = \frac{TN}{TN + FN} \in \mathbb{Q}.$$

All comparisons of these quantities ("Is sensitivity > 0.90?") are decidable in BISH.

Proof. Rational arithmetic is closed under division (with nonzero denominators). Comparison of rationals is decidable by cross-multiplication. □

4.2 The Bayesian Update: Rational vs Real

Proof of Theorem B. The posterior (1) with $\mathrm{sens}, \mathrm{spec} \in \mathbb{Q}$ and $\pi \in \mathbb{R}$ is a composition of rational operations with one real input. Since the function $f(\pi) = \mathrm{sens} \cdot \pi / [\mathrm{sens} \cdot \pi + (1 - \mathrm{spec})(1 - \pi)]$ is a rational function of π, the output is a computable real when π is computable. By Bishop–Bridges [1], §2.3, comparing a computable real with a rational requires either an explicit separation witness (constructive) or Markov's Principle (to guarantee termination of the approximation).

The mechanism is identical to Paper 103's Theorem B: there, $\Phi(z)$ was transcendental for algebraic z (Siegel–Shidlovskii); here, $f(\pi)$ is real-valued for real π. Both require $\mathsf{BISH} + \mathsf{MP}$ for comparison with a rational threshold. □

Proof of Theorem D. When $\pi = \pi_{\mathrm{cohort}} \in \mathbb{Q}$ (e.g., disease prevalence observed in a finite cohort study), the posterior $P(D \mid +) \in \mathbb{Q}$ by closure of $\mathbb{Q}$ under field operations. Comparison with $\tau \in \mathbb{Q}$ is decidable by rational comparison. No omniscience principle is invoked. □

4.3 The Diagnostic Penumbra

Proof of Theorem A. Define the total diagnostic uncertainty $\delta = \delta_{\mathrm{prev}} + \delta_{\mathrm{cv}}$, where δ_{prev} is the half-width of the 95% confidence interval on the prevalence estimate and δ_{cv} is the analytical coefficient of variation of the assay scaled to the posterior.

The posterior $P(D \mid +)$ lies in the penumbra at threshold τ when $|P(D \mid +) - \tau| < \delta$. In this band, the classical value of $P(D \mid +)$ may be above or below τ, but the uncertainty δ prevents constructive certification of which side.

The penumbra has width 2δ, symmetric around τ. This is the diagnostic analogue of the Asymptotic Penumbra (P103), where width $= 2\varepsilon$ with ε the Studentized Berry–Esseen error. □

4.4 ROC Analysis: Continuous vs Discrete

Proof of Theorem C. The area under the ROC curve is $\mathrm{AUC} = \int_0^1 \mathrm{ROC}(t)\,dt$, a Riemann integral of a monotone nondecreasing function $[0,1] \to [0,1]$. Comparing $\mathrm{AUC}_1 > \mathrm{AUC}_2$ for two diagnostic tests is a comparison of two real numbers. By Bishop–Bridges [1], universal real comparison $\forall x, y \in \mathbb{R}\,.\, x < y \vee x = y \vee x > y$ is equivalent to LPO.

The discrete bypass: when the ROC curve is approximated by finitely many operating points (each from a contingency table at a specific threshold), the trapezoidal AUC is a rational number, and comparison is decidable in BISH.

This parallels the exact-vs-asymptotic dichotomy in P103: Fisher's exact test (BISH, zero penumbra) vs asymptotic z-test ($\mathsf{BISH} + \mathsf{MP}$, nonzero penumbra). □

4.5 Treatment Threshold: Rational vs Continuous

When utilities are discrete and rational, the Pauker–Kassirer threshold (2) is rational and the decision is BISH. When utilities are defined over continuous outcome distributions (quality-adjusted life years, survival curves), the threshold becomes the root of an expected-utility equation over $\mathbb{R}$, requiring at least WKL for bounded optimisation.

5 Clinical Test Cases

5.1 High-Sensitivity Troponin: ESC 0/1h Algorithm

The ESC 0/1h algorithm [8, 9] for acute chest pain uses high-sensitivity cardiac troponin I (hs-cTnI, Abbott Architect) with thresholds:

- **Rule-out:** hs-cTnI < 5 ng/L at presentation $\Rightarrow$ discharge (NPV $> 99\%$)
- **Rule-in:** hs-cTnI ≥ 52 ng/L $\Rightarrow$ admit for invasive strategy (PPV $\approx 34\%$)
- **Observe:** 5–52 ng/L $\Rightarrow$ serial troponin at 1 hour, clinical reassessment

Mueller validation cohort. In the international validation study [9] ($N = 7{,}998$):

- Rule-out: $n = 4{,}593$ (57.4%), 30-day MACE: 14 (0.3%)
- Observe: $n = 1{,}972$ (24.7%), 30-day MACE: 141 (7.1%)
- Rule-in: $n = 1{,}433$ (17.9%), 30-day MACE: 484 (33.8%)

The observe zone captures **24.7% of all chest pain presentations** — nearly 1 in 4 patients. The event rate of 7.1% sits within the typical treatment-threshold range for invasive investigation ($\tau \approx 5$–15% depending on utilities). This is the Diagnostic Penumbra: the posterior is too close to the threshold for constructive resolution.

CRM analysis. The contingency table at the 52 ng/L threshold gives:

	MACE	No MACE
hs-cTnI ≥ 52	484 (TP)	949 (FP)
hs-cTnI < 52	155 (FN)	6,410 (TN)

All derived quantities are rational: sensitivity $= 484/639 \approx 75.7\%$, specificity $= 6{,}410/7{,}359 \approx 87.1\%$, PPV $= 484/1{,}433 \approx 33.8\%$. These computations are BISH.

With the cohort prevalence $\pi = 639/7{,}998 \approx 8.0\%$, the Bayesian posterior for a positive test is $P(D \mid +) = 484/1{,}433 \approx 33.8\%$ (rational, BISH-decidable). With a population prevalence estimated from meta-analysis ($\pi \in \mathbb{R}$), the posterior inherits real-valuedness and comparison with τ requires $\mathsf{BISH} + \mathsf{MP}$.

5.2 D-dimer for Pulmonary Embolism

The D-dimer assay at the standard threshold of 500 ng/mL (FEU) has sensitivity $\approx 95\%$ and specificity $\approx 40\%$ for PE [11]. At low pretest probability ($\pi \approx 5\%$, Wells ≤ 4), the posterior probability of PE given a negative D-dimer is $\approx 0.65\%$ — well below any treatment threshold. The decision to exclude PE is BISH-decidable even with real prevalence, because the separation from the threshold is so large that any rational approximation suffices.

Age-adjusted thresholds (age $\times 10$ ng/mL for patients > 50 years) reduce unnecessary imaging by approximately 30% [12] while maintaining the NPV above 97%.

The D-dimer case demonstrates a *collapsed penumbra*: the test's high sensitivity creates such strong posterior separation from the treatment threshold that the grey zone effectively vanishes for the rule-out application. The penumbra persists for intermediate pretest probabilities.

5.3 PSA Screening: The Degenerate Penumbra

Prostate-specific antigen (PSA) at the traditional threshold of 4.0 ng/mL has sensitivity $\approx 27\%$ for all-grade prostate cancer and specificity $\approx 85\%$ [13]. The Thompson PCPT data revealed that 15.2% of men with PSA < 4.0 harbour cancer. With the standard threshold:

$$P(\text{cancer} \mid \text{PSA} \geq 4) \approx 31\%, \qquad P(\text{cancer} \mid \text{PSA} < 4) \approx 17.7\%.$$

Both posteriors fall within the clinical grey zone for biopsy decisions (5–40%). The entire PSA range is a grey zone — a *degenerate penumbra* where no single threshold produces clean constructive separation. This pathology motivated the shift from PSA-based decisions to risk calculators incorporating multiple variables.

6 CRM Audit

6.1 Pipeline Classification

Table 1: CRM pipeline for diagnostic testing

Stage	Component	CRM Level
1	Contingency table (count data)	BISH
2	Test characteristics (sens, spec, PPV, NPV)	BISH
3	Likelihood ratios	BISH
4	Bayesian update (rational prevalence)	BISH
5	Bayesian update (real prevalence)	BISH + MP
6	Treatment threshold (rational utilities)	BISH
7	Treatment threshold (continuous utilities)	WKL
8	Clinical decision (rational posterior & τ)	BISH
9	ROC AUC comparison (continuous)	LPO
10	Test comparison (discrete ROC)	BISH
Total:	**7 BISH + 1 BISH+MP + 1 WKL + 1 LPO**	**70% BISH**

6.2 Comparison with Paper 103

The pipeline decompositions are isomorphic: both have 7 BISH + 1 BISH+MP + 1 WKL + 1 LPO. The 70% BISH rate is not a coincidence but reflects the universal structure of the Archimedean obstruction: most clinical computation is rational arithmetic over finite data; the non-constructive cost enters only at the interface between rational data and real-valued functions.

Table 2: Pipeline isomorphism between P103 and P104

Component	P103 mechanism	P104 mechanism	CRM
Raw data	Sample statistics	Contingency table	BISH
Transcendental gate	Normal CDF Φ	Bayes(π_{real})	BISH + MP
Optimisation	Cox MLE	Utility threshold	WKL
Universal comparison	Real p-value	AUC integral	LPO
BISH fraction	70%	70%	—

6.3 What Descends from Where to Where

- BISH + MP $\to$ BISH: Bayesian inference with real prevalence descends to BISH when prevalence is rational (cohort study). *Excision:* replace population prevalence with finite cohort data.
- WKL $\to$ BISH: Continuous utility threshold descends to BISH when utilities are discrete. *Excision:* use the Pauker–Kassirer rational threshold model.
- LPO $\to$ BISH: Continuous ROC AUC descends to BISH with discrete operating points. *Excision:* trapezoidal rule at finitely many thresholds.

Every non-BISH stage admits a BISH descent via discretisation. The logical cost vanishes when the computation stays within $\mathbb{Q}$.

7 Formal Verification

7.1 Lean 4 Bundle Structure

The formal verification comprises 10 module files:

P104_DiagnosticTesting/ - Module Structure

```
OmnisciencePrinciples.lean    -- LPO, WLPO, MP definitions and implications
ContingencyTable.lean         -- 2x2 table, sens, spec, PPV, NPV
BayesianInference.lean        -- Rational/real posterior, MP requirement
ROCAnalysis.lean              -- AUC as integral, discrete bypass
TreatmentThreshold.lean       -- Pauker-Kassirer, rational/continuous
DiagnosticPenumbra.lean       -- Grey zone characterisation
TroponinCalculations.lean     -- ESC 0/1h algorithm verification
ClinicalTestCases.lean        -- D-dimer, PSA test cases
CRMPipeline.lean              -- Pipeline classification table
Paper104_Assembly.lean         -- Master theorem, axiom documentation
```

7.2 Key Code Snippets

BayesianInference.lean - Discrete Bypass

```
theorem discrete_bypass (sens spec pi_cohort tau : Q)
    (h_pi_pos : 0 < pi_cohort) ... :
    Decidable (bayesPosteriorRat sens spec pi_cohort ... > tau) :=
  inferInstance
```

Bayesian posterior (rational).

The proof is `inferInstance`: Lean's type class inference finds the `DecidableEq` instance for $\mathbb{Q}$ automatically. No axioms invoked.

Pipeline classification.

CRMPipeline.lean - Pipeline Statistics

```
theorem bish_count : bishStages.length = 7 := by native_decide
theorem non_bish_count : nonBishStages.length = 3 := by native_decide
theorem bish_percentage : 100 * bishStages.length / allStages.length = 70
  := by native_decide
```

Troponin verification.

TroponinCalculations.lean - Grey Zone Fraction

```
theorem observe_zone_approx :
    (1972 : Q) / 7998 > 24 / 100 /\ (1972 : Q) / 7998 < 25 / 100 := by
  constructor <;> norm_num
```

7.3 Axiom Inventory

Table 3: Documentary axioms in Paper 104

Axiom	Purpose	Reference
`real_posterior_comparison_requires_MP`	Posterior comparison needs MP	Bishop–Bridges §2.3
`auc_comparison_requires_LPO`	AUC comparison needs LPO	Bishop–Bridges §2.3
`continuous_threshold_exists`	Continuous τ^* existence	Pauker–Kassirer
`continuous_threshold_requires_WKL`	Continuous τ^* needs WKL	Paper 23

7.4 Classical.choice Audit

All theorems over $\mathbb{R}$ show `Classical.choice` in `#print axioms` — this is Lean infrastructure (required for real number arithmetic in Mathlib), not classical mathematical content. The constructive stratification is established by proof content: the main theorem `paper104_main` uses only `inferInstance` (decidable instances), `norm_num` (rational arithmetic), and `native_decide` (finite enumeration). No classical choice principle is invoked as mathematical content.

8 Discussion

8.1 The Archimedean Obstruction in Medicine

Paper 103 identified the Archimedean obstruction in the RCT pipeline: rational test statistics pass through the transcendental normal CDF, producing real p-values whose comparison with rational thresholds requires MP. Paper 104 identifies the same obstruction in diagnostic testing: rational test characteristics pass through Bayes' theorem with real prevalence, producing real posteriors whose comparison with rational treatment thresholds requires MP.

The mechanism is identical in both cases:

$$\text{rational data} \xrightarrow{\text{transcendental function}} \text{real output} \xrightarrow{\text{compare with rational}} \text{decision.}$$

The first arrow is BISH; the comparison requires MP. This is the statistical manifestation of the Archimedean Obstruction Thesis (Paper 98): the non-constructive cost enters through the interface between discrete data and the real continuum.

8.2 Clinical Implications

The Diagnostic Penumbra provides a new quality axis for evaluating diagnostic tests, orthogonal to sensitivity, specificity, and predictive values:

1. **Troponin (ESC 0/1h):** Penumbra captures 24.7% of patients. The observe zone is *logically necessary*, not merely a conservative clinical choice.
2. **D-dimer:** Penumbra collapses for the rule-out application at low pretest probability. High sensitivity creates sufficient posterior separation.
3. **PSA:** Degenerate penumbra — the entire range is a grey zone. Poor sensitivity destroys posterior separation.

8.3 The Discrete Bypass Principle

Every non-BISH stage in the diagnostic pipeline admits a BISH descent via discretisation (Theorem D). This is not an approximation; it is an exact logical downgrade. When the clinician uses a cohort prevalence (rational) rather than a population prevalence (real), the posterior is exactly rational and the decision is exactly BISH-decidable.

The clinical implication: *institutional data is logically cheaper than epidemiological estimates.* A hospital's own ACS prevalence from last year's chest pain presentations gives a rational prevalence that makes the Bayesian chain BISH. A meta-analytic estimate from 50 international studies gives a real-valued parameter that lifts the chain to BISH + MP.

9 Conclusion

The diagnostic testing pipeline decomposes into 10 stages: 7 BISH + 1 BISH+MP + 1 WKL + 1 LPO = 10 stages (70% BISH), isomorphic to the RCT pipeline of Paper 103. The Diagnostic Penumbra — the grey zone where the posterior is too close to the treatment threshold for constructive resolution — is the structural twin of the Asymptotic Penumbra. Both arise from the same mechanism: rational data passing through the continuum.

Applied to troponin, 24.7% of chest pain presentations fall in the penumbra. The ESC 0/1h observe zone is not a clinical convenience but a logical necessity. Applied to D-dimer, the penumbra collapses for rule-out at low pretest probability. Applied to PSA, the penumbra degenerates to cover the entire range, reflecting the test's fundamental inability to separate.

The Clinical Sub-Series has now demonstrated the Archimedean obstruction in two independent clinical domains: RCT statistics (P103) and diagnostic testing (P104). The 70% BISH rate — identical in both — is the signature of the universal architecture: most of medicine is rational arithmetic; the logical cost is concentrated at the interface with the continuum.

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Portions of the Lean formalisation use Mathlib [14], the mathematical library for Lean 4.

This paper follows the format guide for the CRM series [15].

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